

Maximum likelihood estimation

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Logistics and reminders

- + HW 1 due tonight
- + HW 2 released, due next Friday
- + Project part 1 released (choosing a dataset + EDA)
 - + Project will be released in several pieces
 - + Part 1 due in 2 weeks
- + Department seminar coming up on 9/11
 - + 11am in ZSR auditorium
 - + Speaker: Robert Langefeld
 - + Attendance part of class participation grade
 - + If you can't attend in person, can instead watch a seminar on YouTube

Recap

Definition: The *likelihood* $L(Model) = P(Data|Model)$ of a model is the probability of the observed data, given that we assume a certain model and certain values for the parameters that define that model.

Coin example: flip a coin 5 times, with $\pi = P(Heads)$

- + Model: $Y_i \sim Bernoulli(\pi)$, and $\hat{\pi} = 0.9$
- + Data: $y_1, \dots, y_5 = T, T, T, T, H$
- + Likelihood: $L(\hat{\pi}) = P(y_1, \dots, y_5 | \pi = 0.9) = 0.00009$

Recap

Maximum likelihood estimation: pick the parameter estimate that maximizes the likelihood.

Coin example: flip a coin 5 times, with $\pi = P(Heads)$

- + Observed data: T, T, T, T, H
- + Likelihood: $L(\hat{\pi}) = (1 - \hat{\pi})^4(\hat{\pi})$
- + Choose $\hat{\pi}$ to maximize $L(\hat{\pi})$

Recap

Maximum likelihood estimation: pick the parameter estimate that maximizes the likelihood.

Coin example: flip a coin 5 times, with $\pi = P(\text{Heads})$

- + Observed data: T, T, T, T, H
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How do I maximize functions?

Calculus!

Take first derivative,
set = 0

Maximum likelihood estimation with calculus

Maximum likelihood estimation: pick the parameter estimate that maximizes the likelihood.

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Step 1: Write down the likelihood

$$L(\hat{\pi}) = (1 - \hat{\pi})^4(\hat{\pi})$$

$$\log(a^b) = b \log(a)$$

$$\log(ab) = \log(a) + \log(b)$$

Maximum likelihood estimation with calculus

Maximum likelihood estimation: pick the parameter estimate that maximizes the likelihood.

Coin example: flip a coin 5 times, with $\pi = P(\text{Heads})$

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- + Likelihood: $L(\hat{\pi}) = (1 - \hat{\pi})^4 (\hat{\pi})$
- + Choose $\hat{\pi}$ to maximize $L(\hat{\pi})$

Step 2: Take the log

$$\begin{aligned} \log L(\hat{\pi}) &= \log \left((1 - \hat{\pi})^4 (\hat{\pi}) \right) \\ &= \log((1 - \hat{\pi})^4) + \log(\hat{\pi}) \\ &= 4 \log(1 - \hat{\pi}) + \log(\hat{\pi}) \end{aligned}$$

Maximum likelihood estimation with calculus

Maximum likelihood estimation: pick the parameter estimate that maximizes the likelihood.

Coin example: flip a coin 5 times, with $\pi = P(\text{Heads})$

- + Observed data: T, T, T, T, H
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- + Choose $\hat{\pi}$ to maximize $L(\hat{\pi})$

Step 3: maximize

$$\log L(\hat{\pi}) = 4 \log(1 - \hat{\pi}) + \log(\hat{\pi})$$

How do we find where maxima/minima occur for a function?

Take derivative, set $= 0$

Maximum likelihood estimation with calculus

- + Observed data: T, T, T, T, H
- + Likelihood: $L(\hat{\pi}) = (1 - \hat{\pi})^4(\hat{\pi})$
- + Choose $\hat{\pi}$ to maximize $L(\hat{\pi})$

$$\hat{\pi} = \frac{1}{5} = \frac{\# H}{\# \text{flips}}$$

Step 3: maximize -- take first derivative and set equal to 0

$$\log L(\hat{\pi}) = \underbrace{4 \log(1 - \hat{\pi})} + \underbrace{\log(\hat{\pi})}$$

$$\frac{d}{d\hat{\pi}} \log L(\hat{\pi}) = \frac{-4}{1 - \hat{\pi}} + \frac{1}{\hat{\pi}} \stackrel{\text{set}}{=} 0$$

$$\Rightarrow \frac{1}{\hat{\pi}} = \frac{4}{1 - \hat{\pi}} \Rightarrow \frac{1 - \hat{\pi}}{\hat{\pi}} = \frac{4}{1}$$

$$\Rightarrow 1 - \hat{\pi} = 4\hat{\pi} \Rightarrow 1 = 4\hat{\pi} + \hat{\pi} \Rightarrow 1 = 5\hat{\pi}$$

$$\Rightarrow \boxed{\hat{\pi} = \frac{1}{5}} \leftarrow \text{maximum likelihood estimate!}$$

Class activity

Work on the activity (handout) with a neighbor, then we will discuss as a class

Class activity

$$+ P(Y_i = 0) = \pi_0$$

$$+ P(Y_i = -1) = 2\pi_0$$

$$+ P(Y_i = 1) = 1 - 3\pi_0$$

Observe data $-1, -1, 0, 1, 0, -1$.

$$L(\hat{\pi}_0) = ?$$

$$= P(-1)^3 P(0)^2 P(1)^1$$

$$= (2\hat{\pi}_0)^3 \hat{\pi}_0^2 (1-3\hat{\pi}_0)$$

$$= 8\hat{\pi}_0^3 \hat{\pi}_0^2 (1-3\hat{\pi}_0)$$

$$= 8\hat{\pi}_0^5 (1-3\hat{\pi}_0)$$

$$L(\hat{\pi}_0) = 8 \hat{\pi}_0^5 (1 - 3\hat{\pi}_0)$$

$$\begin{aligned} \log L(\hat{\pi}_0) &= \log(8 \hat{\pi}_0^5 (1 - 3\hat{\pi}_0)) \\ &= \log(8) + \log(\hat{\pi}_0^5) + \log(1 - 3\hat{\pi}_0) \\ &= \underbrace{\log(8)} + 5 \log(\hat{\pi}_0) + \log(1 - 3\hat{\pi}_0) \end{aligned}$$

$$\begin{aligned} \frac{\partial}{\partial \hat{\pi}_0} \log L(\hat{\pi}_0) &= 0 + \frac{5}{\hat{\pi}_0} + \frac{1}{1 - 3\hat{\pi}_0} (-3) \\ &= \frac{5}{\hat{\pi}_0} + \frac{-3}{1 - 3\hat{\pi}_0} \stackrel{\text{set}}{=} 0 \end{aligned}$$

$$\Rightarrow \frac{5}{\hat{\pi}_0} = \frac{3}{1 - 3\hat{\pi}_0} \Rightarrow \frac{1 - 3\hat{\pi}_0}{\hat{\pi}_0} = \frac{3}{5}$$

$$\Rightarrow 5 - 15\hat{\pi}_0 = 3\hat{\pi}_0$$

$$\Rightarrow 5 = 18\hat{\pi}_0$$

$\Rightarrow$

$$\boxed{\hat{\pi}_0 = \frac{5}{18}}$$

maximum likelihood estimate!

Class activity, Part 2

Maximizing likelihood for a logistic regression model in R:

https://sta214-f25.github.io/class_activities/ca_06_2.html

(You do not need to submit this part of the activity)